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Investigating intersemiosis: a systemic functional multimodal discourse analysis of the relationship between language and gesture in classroom discourse

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ABSTRACT

A challenge for researchers working with multimodal classroom discourse is to be able to describe and discuss the interaction and interplay across various semiotic resources. This article adopts the Systemic Functional Multimodal Discourse Analysis approach to examine the relationship between language and gesture used by the teacher, in interaction with the students, and the emergent meanings made multimodally. It discusses the mechanisms by which language and gesture combine to make meaning by extending concepts originally developed for language–image relations. From the analysis and interpretation of the teachers' multimodal selections in the lesson, the emergent meaning of 'structured informality' is proposed. Structured informality offers a way to consider how a teacher can design an effective learning experience for their students using multimodal resources.

KEYWORDS

classroom discourse • gesture • intersemiosis • language–gesture relations • multimodal discourse analysis • systemic functional theory

MULTIMODAL CLASSROOM DISCOURSE

Multimodality involves 'how different kinds of meaning making are combined into an integrated, multimodal whole' (Jewitt et al., 2016: 2). The multimodal approach to classroom discourse is the investigation of a range of semiotic resources, beyond just spoken and written language alone. Specifically, multimodality focuses on exploring the interactions and interplay across the semiotic resources in the constellation of meanings made. As Iedema (2003: 31)

Visual Communication 2019
Vol. 0(0) 1–25
© The Author(s) 2019
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DOI 10.1177/1470357218820695
SAGE

argues 'semiosis is not analysed in terms of discrete building blocks or structures, but in terms of socially meaningful tensions and oppositions which could be instantiated in one or more ways.'

Past research into classroom discourse and interaction between teacher and students has mostly tended to focus on the analysis of classroom spoken and written language alone (see, for example, Mercer, 2000; Sinclair and Coulthard, 1975; Walsh, 2006). The term 'language' follows its conventional usage in linguistics as spoken and written discourse, although, in the classroom corpus used in this article, it refers specifically to teacher-talk in the English language. The concentration on just the linguistic aspect of classroom interaction does not account for the combination of semiotic resources that together, rather than separately, make the teaching and learning experience. Kress (2003: 35) argues that 'language alone cannot give us access to the meanings of the multimodally constituted messages; language and literacy now have to be seen as partial bearers of meaning only'. O'Halloran (2007: 79) also observes that 'the study of linguistic discourse alone has theoretical limitations which have the potential to simplify and distort the actual nature of pedagogical practice'. Hence, educational research can benefit from moving beyond a study on language to examine the other multimodal resources used in the classroom.

From a multimodal perspective, meanings are made in the classroom through the teacher's use of language, gesture, space and positioning, as well as resources such as textbooks, whiteboard and educational technology. Kress et al. (2005) demonstrate that, even in an English-language classroom, language may not always be the dominant semiotic resource. In a similar vein, Bourne and Jewitt (2003) investigate the ways in which the interpretation of a literary text is mediated and (re)construed through social interactions. Their work demonstrates how a multimodal analysis of classroom discourse can yield a richer understanding of the teaching and learning in the classroom. Advances have also been made in the study of the teachers' use of multimodal resources across various subject classrooms, such as Moschkovich (2002) and Greiffenhagen (2008) in the Mathematics classroom, as well as Kress et al. (2001) and Roth and Jornet (2014) in the Science classroom. With this recognition, and the ease, through the use of technology, of the collection and analysis of multimodal classroom data in educational research today, an increasing number of scholars in the field of multimodality have applied the multimodal lens to look at meaning-making in the classroom. (see, for example, Jewitt, 2013; O'Halloran, forthcoming; Takahashi and Yu, 2017; Taylor, 2014). A continuing challenge for researchers working with multimodal discourse is to develop the theoretical apparatus for describing and discussing the interaction and interplay across various semiotic resources used and the emergent meanings made multimodally. While recognizing the range of semiotic resources used by the teacher in meaning-making, this article focuses on the relationship between language and gesture by drawing from a corpus of multimodal classroom discourse data of two teachers in Singapore.

The aim of this article is to study the teachers' use of language and gesture from the Systemic Functional Multimodal Discourse Analysis (SFMDA) approach (O'Halloran and Lim, 2014) and propose the relationships and mechanisms in which language and gesture combine to make meaning by extending concepts originally developed for language–image relations. The article also discusses the notion of 'structured informality', which offers a way for teachers to consider how to orchestrate the multimodal resources in the classroom to design an effective learning experience for their students.

SYSTEMIC FUNCTIONAL MULTIMODAL DISCOURSE ANALYSIS APPROACH

One of the approaches to study multimodality is informed by the work in linguistics, in particular, of Michael Halliday's Systemic Functional Theory. Halliday (1978, 1994[1985]) conceives language as a social semiotic system and has developed a robust theoretical apparatus, the systemic functional grammar, to systematize the meaning potential of language and foreground the notion of choice in meaning making. The extension of Systemic Functional Theory as a productive approach to investigate other semiotic resources is pioneered by O'Toole (2010[1994]) and Kress and Van Leeuwen's (2006[1996]) influential work on images. Others, such as Hood (2007, 2011), Martinec (2000, 2001, 2004), Zappavigna et al. (2010), Cleirigh (2011) and Lim (2010, 2017), have also applied Systemic Functional Theory as an approach to systematize gestures.

This article builds on the work of the researchers within the Systemic Functional Theory tradition and adopts the SFMDA approach to investigate the ways in which language and gesture work together in intersemiosis by extending the concepts developed for language–image relations to that of language and gesture. The SFMDA approach focuses on the 'grammatics' of semiotic resources in fulfilling specific functions and how the semiotic choices work in combination to make meaning over space and time (O'Halloran and Lim, 2014). A key aspect in the SFMDA approach is its study of 'multimodal grammatics, where interacting systems of meaning are a key motif' (Jewitt et al., 2016: 39). Processes and mechanisms of intersemiosis are of particular interest and are central to the approach. As such, this article will discuss the ways in which language and gesture work together to produce a coherent and cohesive multimodal classroom discourse.

Intersemiosis

Given that classroom discourse is multimodal in nature, it is useful to investigate how the teacher's use of various semiotic resources work together to make meaning in the classroom. O'Halloran (2005: 159) proposes the term 'intersemiosis' to describe 'the meaning arising across semiotic choices'. Intersemiosis has also been observed by Royce (1998, 2006) as 'intersemiotic complementarity' where 'visual and verbal modes semantically complement

each other to produce a single textual phenomenon' (Royce, 1998: 26). Lemke (1998) notably describes the meanings arising from the combination of semiotic resources in scientific discourse in textbooks as a 'multiplying of meanings'. Hence, the 'emergent meaning' (Lim, 2004) arising from the combined use of the semiotic resources is more profound than the meaning made by an individual semiotic resource. While this is generally recognized, what is of interest in intersemiosis are the ways, or mechanisms, in which the semiotic resources work in combination to make meaning. This is discussed in relation to the teacher's use of language and gesture in this article.

Thibault (2004) explains that language and gesture are very different semiotic resources and are organized according to different principles. He (2004: 26) observes that 'language is predominantly typological-categorical; it is based on discrete categorical contrast or difference. Gesture, on the other hand, is topological-continuous; it is based on continuous variation of visual and spatial relations.' As such, language and gesture 'do not simply express the same meanings by alternative means of expression. Instead, they make different meanings on the basis of their different principles of organization.'

Baldry and Thibault (2006: 18) propose the 'Resource Integration Principle', where the semiotic resources in a multimodal texts are not 'simply juxtaposed as separate modes of meaning making but are combined and integrated to form a complex whole which cannot be reduced to, or explained in terms of the mere sum of its separate parts'. While the Resource Integration Principle is in operation within all multimodal texts, Baldry and Thibault (2006: 4) caution that 'different modalities adopt different organisational principles for creating meaning.' Hence, it is necessary to examine the specific semiotic resources in focus within the multimodal text and explore the unique ways in which they combine and interact in their joint co-deployment.

Matthiessen (2009) also describes the orchestration of multimodal semiotic resources as 'semiotic harmony'. Matthiessen (2009: 11) observes that 'one interesting – and critical – aspect of the division of semiotic labour among the denotative semiotic systems is the extent to which they operate in semiotic harmony with one another.' Nonetheless, like Baldry and Thibault (2006), Matthiessen (2009) acknowledges that how this is accomplished is unique to different resources. Specifically of interest is 'how the different resources for creating meaning complement one another and how the semiotic labour (the work of creating meaning in context) is divided among them' (p. 23). In this light, this article investigates the specific ways in which the semiotic resources of language and gesture interact and complement one another in semiotic harmony.

The intersemiosis between the semiotic resources of language and images has been most theorized to date. Systems and theories have been proposed by many researchers to investigate language–image relations (see, for example, Bateman, 2014; Daly and Unsworth, 2011; Lim and O'Halloran, 2012; O'Halloran and Lim, 2009; Painter et al., 2011; Royce, 2006). In particular,

Unsworth and Cleirigh (2009) argue that 'advancing understanding of how images and language interact to construct meaning seems crucial in seeking to reconceptualize literacy and literacy pedagogy from a multimodal perspective'. This article advances the same argument that understanding how language and gesture are used by the teacher in meaning-making is critical in helping us reconceptualize pedagogy.

GESTURE STUDIES

Gesture, as a topic, has been well studied across diverse fields spanning from Ekman and Wallace (1969, 1974), Goldin-Meadow and Singer (2003) and McNeill (1992, 2005) in psychology, Kendon (1996, 2004) and Haviland (2004) in anthropology, and Roth (2001) and Cienki (2008) in cognition and the learning sciences. Within linguistics, work on gesture, from the conversation analysis perspective, is represented in the seminal work of Goodwin (1979) and Shegloff (1984), and Ochs et al. (1996).

Scholars in the field of conversation analysis have also argued that gesture should be studied together with language, rather than examined in isolation. For instance, Goodwin (2000: 1499) observes that gesture and language 'mutually elaborate each other' and that gestures can 'function as individual actions, or as components of multimodal action' (p. 1519). Mondada (2016: 338) observes the extension of interests, within the field of conversation analysis, beyond verbal resources to 'also embodied ones, within a more global and holistic approach of multimodality, comprising language, gesture, gaze, head movements, facial expressions, body posture, body movements, and embodied manipulations of material objects. This was described by Nevile (2015) as an 'embodied turn'. The interest in research on multimodality paves ways in 'which the multimodal dimension of understanding can be empirically and analytically considered' (Mondada, 2011: 550).

The different approaches in the study of gesture as well as on language and gesture foreground specific distinctive features in the use and co-deployment of these resources. Many of the approaches, such as the multimodal (inter)action analysis (Norris, 2004, 2016), and conversation analysis approach to embodied interactions (Broth and Mondada, 2013; Streeck et al., 2011), explore the orchestration in the range of semiotic resources in communication at the 'multimodal semantics' level. There is generally a lesser focus on the analysis of the specific relationships and mechanisms by which the semiotic resources interact and interplay at the granular level in specific instantiations. An investigation of the 'multimodal grammatics' offered through the SFMDA lens can provide insights into how the systems of meaning interact to produce a coherent emergent multimodal discourse.

Working within the Systemic Functional Theory tradition, Martinec (2000: 243) organizes the classifications of actions as Presenting Action, Representing Action and Indexical Action. Presenting Action 'communicates

non-representational meanings' and consists of actions such as scratching, lifting an object, flipping a page. They often serve a practical function. Representing Action is gesture with a conventional signifying function. They 'function as a means of representation' and are strongly coded representations. In addition, Representing Actions in contextualizing relationship with language also makes meanings beyond just signalling the status of the accompanying linguistic text. Lim (2017) further distinguishes Language Correspondent Gestures or Language Independent Gestures in Representing Action. Indexical Action necessarily accompanies language and requires it for interpretation. However, there are other types of gestures and other meanings made in gesture as well. Indexical Action usually co-occurs with speech and 'in order to retrieve its full meaning, one has to have access to the second-order context which is represented simultaneously in indexical action and concurrent speech' (Martinec, 2000: 244). Indexical Action are Language Dependent Gestures (Lim, 2017).

When a combination of semiotic selections is co-deployed, the co-instantiation in time and the co-occurrence in space of these resources contribute to intersemiosis. While both temporal co-instantiation and spatial co-occurrence are fundamental in intersemiosis, spatial co-occurrences are foregrounded in static texts, such as print advertisements and textbooks, whereas temporal co-instantiations are foregrounded in dynamic texts, such as in film texts and in the environment of multimodal classroom discourse. Spatial co-occurrence of image and language within the same shared space on a page results in intersemiosis. Temporal co-instantiation operates on different time scales – minutes, seconds and micro-seconds, following Lemke's (2000) multiple time scales. Temporal co-instantiation of resources, such as language and gesture happening in the same timescale, results in intersemiosis.

The application of concepts developed in the intersemiosis of language and images to language and gestures is performed circumspectly. This is because most of the intersemiotic mechanisms theorized for language–image relations are based on spatial co-occurrences of images and language on the same page, whereas language–gesture relations are based primarily on their temporal co-instantiations. Hence, not all the conceptions proposed for language–image relations can always be mapped to that of language and gestures. While this is the case, the concepts may not be fully incompatible as well. Within the SFMDA approach to multimodal discourse, the intersemiosis between language and images is by far the most well studied. This article explores the usefulness of the ideas developed for language–image relations in providing a language to describe and discuss the relationship between language and gesture used by the teacher in the classroom. The productive extension of the intersemiotic systems between language and images to that of language and gestures, and subsequently to other semiotic resources, can point towards the possibility of a unifying theory of intersemiotic organization across all semiotic resources.

Contextualizing relations

Intersemiosis between language and gesture arises from the contextualizing relationships, which are the meanings made from the spatial co-occurrences or temporal co-instantiations of the two semiotic resources. As Thibault (2000: 362) explains, it is 'on the basis of co-contextualizing relations that meaning is created'. Lim (2004: 239) distinguishes two types of contextualizing relations:

> In cases where the meaning of one modality seems to 'reflect' the meaning of the other through some type of convergence, the two resources share co-contextualizing relations. On the other hand, in cases where the meaning of one modality seems to be at odds with or unrelated to the other, their semantic relationship is one that creates divergence or dissonance. In the latter case, the resources share re-contextualizing relations.

The intersemiosis between language and images has been studied from the perspective of contextualizing relationships (Lim and O'Halloran, 2009; O'Halloran et al., 2016). In examining the contextualizing relationships in mathematical discourse, O'Halloran (2007: 95) discusses how co-contextualizing relations and re-contextualizing relations create 'new semantic layers' and enable 'new … content to be foregrounded'. Re-contextualizing relations also creates new semantic layers. However, this is realized through the reconciliation of the divergent meanings made. This emergent meaning usually carries new semantic layers, such as irony, sarcasm, dilemma and ambivalence. For communication to be successful, both semantic convergence and divergence through the combination of semiotic selections must be reconciled and ultimately serve the intent of the message. Re-contextualizing relations leading to a semantic divergence that cannot be reconciled result in ambiguity, confusion and a consequent breakdown in communication.

Semiotic cohesion

In describing language–image relations, O'Halloran (2008: 453) proposes the notion of Semiotic Cohesion, where 'system choices function to make the text cohesive'. The conception of Semiotic Cohesion is extended by Liu and O'Halloran (2009: 367) with the proposal of 'intersemiotic texture' in 'a coherent multimodal message'. Liu and O'Halloran (2009) argue that Semiotic Cohesion is achieved through the mechanisms of Intersemiotic Parallelism and Intersemiotic Polysemy to account for the meanings made through Semiotic Cohesion.

Following from Hasan's (1985) work on parallelism in the texture of language, Intersemiotic Parallelism refers 'to a cohesive relation which interconnects both language and images when the two semiotic components share a similar form' (Liu and O'Halloran, 2009: 372). Intersemiotic Polysemy refers 'to the cohesive relation between verbal and visual components, which share multiple related meanings in multisemiotic texts' (Liu and O'Halloran, 2009:

375). This is based on Halliday and Hasan's (1985) Lexical Cohesive Relations and follows from Royce's (1998) Ideational Intersemiotic Complementarity. Liu and O'Halloran (2009: 375) explain that 'Intersemiotic Polysemy results in co-contextualization relations between language and images, and experiential convergence in multi-semiotic texts.' While originally formulated to investigate language–image relations in chemistry discourse, Liu and O'Halloran's (2009) propositions are extended in this study to describe the relationships between language and gesture used by teachers in the classroom. The following section presents a brief review of the teachers' choices in language and their use of gestures in the larger study so as to provide a context for the discussion on the emergent meanings made in the intersemiosis.

AUTHENTIC CLASSROOM CORPUS

The discussion on the relationship between language and gesture in this article draws examples from a corpus of authentic classroom data collected from two teachers, teaching in a pre-university setting in Singapore. This investigation of the intersemiosis of language and gesture is part of a larger study on the two teachers' orchestration of multimodal resources to design their students' learning experience in the context of a Singapore school.

With permission from the school principal and the teacher-participants, the teachers' language, gestures, and their use of space and positioning in specific lessons were recorded, annotated and analysed. The multimodal data corpus was collected through the use of closed-circuit cameras that had been mounted on the ceilings of the classroom. While the multimodal data across the various semiotic resources have been analysed in the larger study, this article focuses on the discussion on the teachers' use of language and gesture in one of the lessons.

The two teacher-participants, given the pseudonyms of Andy and Betty, differ in gender and also in teaching experience. Andy, male, in his 20s, is a novice teacher with less than two years of experience. Betty, female, in her 30s, is an experienced teacher and has taught for more than 10 years. It is thus of interest in the larger study to observe how the teachers' use of multimodal resources can design different learning experiences for their students.

Discussions of the linguistic analysis and gestural analysis are presented in the sections below. A linguistic analysis and gestural analysis of the meanings were carried out through the lens of Systemic Functional Theory as part of the larger study. As the focus of this article is on the relationship between language and gesture, only a summary and selection of the linguistic analysis is described. This is to provide a sense of how Andy and Betty have used language in their lessons. Thereafter, instances of contextualizing relations between language and gesture are discussed through the lens of semiotic cohesion. The emergent meaning of structured informality, as a way that teachers orchestrate various modalities to design effective learning experience for students, is proposed based on the multimodal analysis from the larger study.

LINGUISTIC ANALYSIS

In Systemic Functional Grammar, linguistic choices in the mood system realize interpersonal meanings, that is, to enact social relationships with language. In terms of the mood system, both Andy and Betty typically use more finite mood choices. The frequent use of modality represents a modulation of power through the use of low modality such as in words like 'could' and 'might' and an assertion of power through the use of high modality such as in words like 'should' and 'must'. Words that indicate power, such as 'don't' and 'need' are used very frequently by Andy and to a lesser extent by Betty. The frequent use of high modality, coupled by the frequent use of imperatives represented in the non-finite selections, suggests more instances where Andy asserts his authority during the lesson.

Linguistic choices in the transitivity system realize ideational meanings, that is to construct our experience of the world with language. Both Andy and Betty use mostly relational processes in the classroom. Halliday Halliday (1994[1985]: 119) explains that 'relational processes are a process of being … a relation is being set up between two separate entities.' Relational clauses are also a regular feature in pedagogic discourse as they contribute to the scaffolding of understanding and knowledge construction. The category of relational process is represented in the sub-classification of Relational-Attributive and Relational-Identifying. Martin et al. (1997: 106) explain that 'the fundamental difference between attributive and identifying is the difference between class membership (attributive) and symbolization (identifying)'. Andy uses more relational-identifying clauses whereas Betty uses more relational-attributive clauses. Andy uses more Relational-Identifying, as compared to Betty. For instance, Andy tends to formulate Relational-Identifying clauses such as 'That is one of the points' and 'This is not in the context of our discussion'. These are examples of Andy's preference towards relating two different levels of abstraction – token and value, symbolically. Betty uses more Relational-Attributive clauses in comparison to Andy. For instance, Betty tends to use Relational-Attributive clauses such as, 'We have three points' and 'They are too self-centred'. These are examples of Betty's preference towards relating 'carrier and attribute, of the same level of abstraction, but differ[ing] in generality as member to class, subtype to type' (Martin et al., 1997: 106). In the first comparison, the choice made by Betty using the Relational-Attributive process of 'We have three points' creates a sense of solidarity and shared ownership of learning with the students. This contrasts with Andy's more detached and formal deictic references of 'That is one of the points' and 'This is not in the context of our discussion'. In the second comparison, a more direct comparison is made between 'they' and 'self-centred' through the Relational-Attributive process. This contrasts with an equivalent recast into the Relational-Identifying process of 'They are self-centred people.'

The 'clauses of feeling, thinking and perceiving' are described by Halliday (1994[1985]: 114) as the mental process. These clauses are sub-classified as Mental-Affective, Mental-Cognition and Mental-Perception. Given the nature of teaching and learning, it is unsurprising that these processes are featured prominently in Andy's and Betty's lessons. Andy also uses more mental-cognitive processes as opposed to Betty's usage of mental-affective processes. Andy tends to represent entities through the cognitive processes, whereas Betty evokes greater affect in her linguistic choices. The high amount of these processes are a result of Andy's tendency to use the words 'think' and 'look', respectively. Significantly, Mental-Affective is the only type of mental process that Betty has almost twice as much as Andy. This is a result of Betty's more regular use of the word 'feel'. For example, in her question, 'Do you feel it will worsen?' as opposed to Andy's 'Do you think this problem can be resolved?' The use of the Mental-Affective invokes an emotional element into the question and encourages the students to engage with the issue, not just cognitively but also emotionally. The connection with issues on a personal and emotional level, as well as on the intellectual level, is tacitly encouraged through the use of the Mental-Affective process.

Behavioural and Existential processes are used very rarely by both teachers. Betty seems to have slightly more of these processes than Andy. Behavioural processes 'construe human behavior' (Martin et al., 1997: 109). Arguably, the presence of behavioural processes such as in Betty's questions of 'Why are you smiling' and later 'Why are you laughing', convey an affective dimension to the professional nature of teacher–students interaction.

From the linguistic analysis, it is observed that Andy uses more imperatives and high modality to indicate his power and authority. In contrast, Betty prefers adjuncts and low modality to build solidarity with the students. It is also observed that Betty has more occurrences of students' laughter during her lesson. This suggests an ease between the students and her as well as their enjoyment of the lesson.

GESTURAL ANALYSIS

From the analysis of the teachers' gestures in the larger study, it is observed that Andy uses more Presenting Actions (Martinec, 2000) than Betty. As described earlier, Presenting Actions do not have any signifying function. While they convey a sense of dynamism in the lesson, they can also be distracting if used excessively. In comparison, Betty uses more Indexical Actions and Representing Actions. These actions serve a signifying function as Language Correspondent Gesture or Language Independent Gesture (Lim, 2017).

It has been observed that Andy tends to rest his hands by the side of his legs whereas Betty tends to hold her hands at the waist level. The latter conveys a sense of readiness and professionalism. Textually, the gestures made by Andy are usually fast and often of a single beat. In comparison, Betty makes her gestures more

slowly and almost deliberately to advance her pedagogic point. She is observed to do this regularly with the aid of rhythmic beats for emphasis.

Students are usually the directional goals to whom Andy points to with his hand. This usually draws the students' prompt attention. However, the act of pointing at students is also an exercise of authority and power that Andy carries as a teacher. This is despite his deictic gesture being moderated frequently by the option to point more generally with the hand, rather than with a finger. In contrast, Betty usually points to the screen with her index finger and often through the visualizer. This suggests precision and focus on the lesson materials projected on the screen. From the gestural analysis, it has been observed that many of Andy's choices in gestures are more arbitrary and polarized – ranging from casualness to authoritativeness. In comparison, Betty uses her gestures quite deliberately to convey a formality and professionalism, while maintaining a sense of openness and invitation.

INTERSEMIOSIS IN LANGUAGE AND GESTURE

Selected temporal co-instantiations of language and gesture used by the teachers are identified to discuss the mechanisms responsible for the intersemiosis between them. The examples are identified to examine the applicability and productivity of concepts developed for language–image relations, to that of language and gesture. Specifically, the concepts of contextualizing relations and the mechanisms of Intersemiotic Parallelism and Intersemiotic Polysemy to realize semiotic cohesion are investigated.

Indexical action and language

As mentioned earlier, Indexical Actions are language dependent gestures where the meanings made in gesture are accessed through the meanings made in the accompanying language (Lim, 2017). The co-contextualizing relations between Indexical Action and language result in semantic convergence. However, while there is a mutually reinforcing effect in the emergent meaning, it is often not simply a repetition of the same meaning made in language. There can be a variety of meanings which Indexical Actions can add to the semantics in language. For instance, in the observed lessons of Betty and Andy, many of the Indexical Actions realize the meanings of Importance, Receptivity and Relation. The three major types of realizations of ideational meanings were identified as being frequently occurring in the multimodal classroom discourse from the larger study and are described in greater detail in Lim (2017). The representation of Importance is instantiated through rhythmic beat, the representation of Receptivity is instantiated through open palms, and the representation of Relation is instantiated through pointing. In its temporal co-instantiation with language, Intersemiotic Polysemy operates to construct an additional layer of semantics, an emergent meaning, absent from the use of language alone.

Figure 1. Indexical Action realizing Importance.

For instance, the meaning of Importance is realized in the rhythmic beats of the arm (see Figure 1). For example, when Andy draws attention to the focus of the class discussion, he states, 'What we are talking here about is accommodating bio-fuel.' The significance of this clause in orientating the discussion is accompanied by four rhythmic beats in Indexical Action realizing the representation of Importance. Likewise, Indexical Actions realizing the representation of Importance are used regularly by Betty to highlight the various important points in her lesson. For instance, this is observed when she reminds the students what to avoid stylistically in their answers to the Application Question. She remarks, 'That is very ugly, very unprofessional. We don't do that.' Her speech is co-instantiated with her gesture of six rhythmic beats of the right arm which realizes the meaning of Importance.

The rhythmic beats in themselves do not produce semantics. However, Intersemiotic Polysemy, operating in the co-contextualizing relations between language and gesture, brings about semantic convergence. As such, the meaning of Importance realized by the gestural selections converges and reinforces the status of the meanings made in language.

The gestural meaning of Receptivity in Indexical Actions usually accompanies questions where a response from the student is invited (see Figure 2). This is observed when Andy asks a student, 'So that is first [what] we have to figure out, isn't it?' The Indexical Action instantiated by the outstretched open palms comes into co-contextualizing relations with the linguistic text. As the gesture realizes openness and invitation, Intersemiotic Polysemy operates to reinforce the question and the invitation for participation from the students.

The co-instantiation of questions with Indexical Actions realizing the meaning of Receptivity are also observed in Betty's lesson, Betty uses this gesture frequently to accompany statements as well (see Figure 2). For instance, this is observed when she says, 'In a bit, when I go through the answers with you again, I will show you how it is done.' She realizes the meaning of Receptivity

Figure 2. Indexical Action realizing Receptivity.

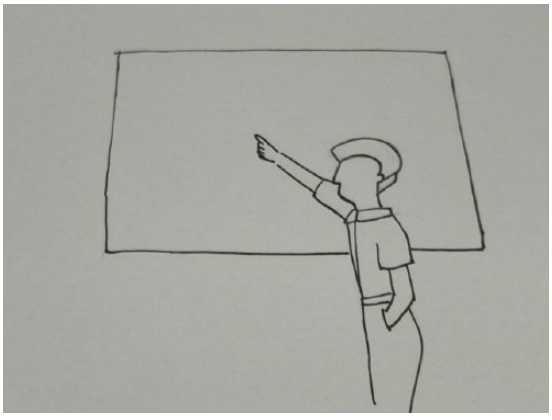

Figure 3. Indexical Action realizing Relation.

with her open palms and conveys welcome and invitation. Intersemiotic Polysemy results in semantic convergence. Through it, a sense of openness and offer is added to the meaning made in the statement.

The meaning of Relation in Indexical Action is realized through the act of pointing (see Figure 3). For instance, this is observed when Andy points at a student and asks the question, 'What do we have to do?' The question, while it could be interpreted as addressing the entire class, is shown to be directed to a specific student through its co-instantiation with the gesture. His gestural selection realizes the meaning of Relation. It also comes into co-contextualizing relations with the question. Intersemiotic Polysemy operates to produce specificity in the emergent meaning.

Similarly, in another instance, Andy points at the whiteboard as he commands, 'Pay attention here OK.' The linguistic text alone does not provide enough information on where the 'here' refers. However, Intersemiotic

Figure 4. Indexical Action realizing Defensiveness and Disapproval.

Polysemy disambiguates speech in its co-contextualizing relations with the act of pointing. The deictic gesture indicates the 'here', in this case, the whiteboard, as the goal of where the students are to pay attention. As such, the combination of semiotic selections in language and gesture results in semantic convergence and produces a coherent emergent meaning for the students.

In addition to co-contextualizing relations, Indexical Actions can also enter into re-contextualizing relations with language. An instance of this is observed in Betty's Indexical Action of folding her arms across her chest (see Figure 4). Interpreted as an Indexical Action, rather than a Presenting Action to warm herself, the action can realize the meanings of defensiveness. In this instance, this action is co-instantiated with her linguistic anecdote of a former student who asked her out on a date. She says, 'I told you about the boy who messaged me right, after he graduated. My husband should never see this, otherwise he won't sleep anymore.' While the telling of the anecdote conveys openness and represents an invitation into her personal life, this is re-contextualized with her gestural selection of folded arms. The semantic divergence can be reconciled through Intersemiotic Polysemy where her gesture is construed as a delineation of professional boundary as represented by the physical barriers of the arms, even as she metaphorically reaches out to the students through language.

Representing Action and language

Representing Action is categorized as Language Independent Gesture and Language Correspondent Gesture (Lim, 2017). Language Independent Gestures can make meaning on their own without language, although they may also be accompanied by language. Language Correspondent Gestures are gestures that co-occur with speech, but their meanings can be accessed and interpreted without relying on the accompanying language.

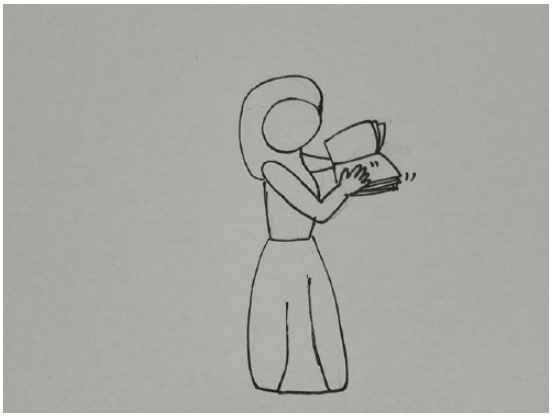

Figure 5. Flipping Action.

Representing Action can also come into either co-contextualizing relations or re-contextualizing relations with language. In contrast with the intersemiosis between Indexical Action and language, which usually relies on the operations of Intersemiotic Polysemy, Intersemiotic Parallelism is in operation for some Language Correspondent Gestures. Intersemiotic Parallelism results in semantic convergence.

Intersemiotic Parallelism is in operation when similar meanings are made both in language and in gestures. For instance, Betty makes the action of flipping with her right hand when she says, 'You flip through a magazine' (see Figure 5). The temporal co-instantiations of the linguistic and gestural selections reinforce the meanings made through the visualization of the linguistic process 'flip' in the Language Correspondent Gesture. Similarly, this is also observed when Betty reminds the students, 'Don't quote the whole chunk of four, five lines' in their answer to the Application Question. Her verbal text 'Don't' co-instantiates with the shaking of both hands to signify negation (see Figure 6). Intersemiotic Parallelism operates in the co-contextualizing relations between language and gesture in both examples to achieve semantic convergence. This, in turn, enhances the emergent meaning made in the combinational deployment. In the former, it accentuates the process of 'flip', which is what Betty wants the students to do. In the latter, the visual gestural repetition of 'Don't' emphasizes what they should not do in their answers.

Likewise, Intersemiotic Parallelism is in operation when Andy describes the new template he is proposing as 'This is linear and downward', as his words co-instantiate with a swift downward movement of his left hand (see Figure 7). The Representing Action in the downward movement of the hand comes into co-contextualizing relations with his words. Semantic convergence occurs as the meanings made in language are reinforced in the gestural realization.

Continuing with his description of the template, Andy makes a Language Independent Gesture in Representing Action. This is observed

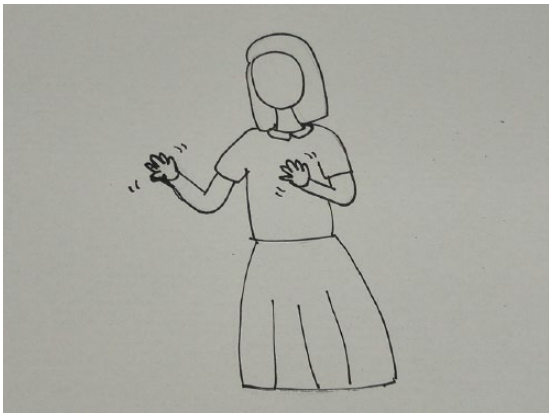

Figure 6. Shaking of Hands to Signify 'Don't'.

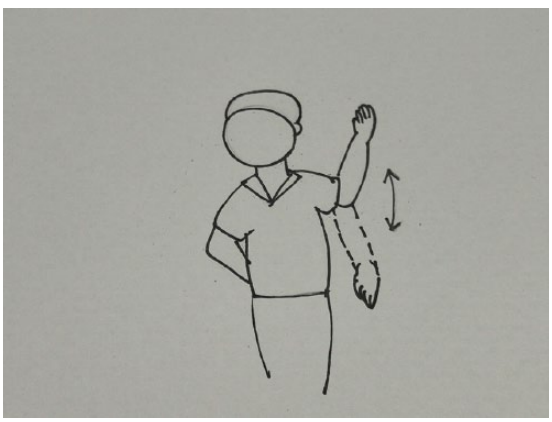

Figure 7. Swift Downward Movement of Hand.

when he stops before he completes his sentence as he says, 'and this is like …' However, the gap in Andy's speech is filled by the sideward movement of the left hand to signify a horizontal line (see Figure 8). The co-contextualizing relations between the Language Independent Gesture and the speech produce semantic convergence. The emergent meaning is the understanding that the template has a horizontal line across it. In this case, the Language Independent Gesture is used to fill in insufficient or missing information from language. The operations of Intersemiotic Polysemy realizes a coherent emergent meaning in the combination of semiotic selections.

The meanings in Language Correspondent Gesture can be understood without relying on the accompanying language and may not replicate the same meanings in language. For instance, this is observed when Betty makes a swift chopping gesture with her right hand as she says, 'Now those of you who score below three, evaluation is missing from your

Figure 8. Sideward Movement of Hand.

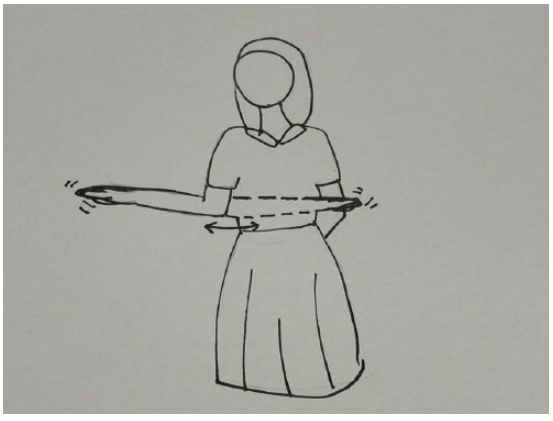

Figure 9. Swift Chopping Action.

answer' (see Figure 9). In its co-instantiation with the linguistic text, the chopping action realizes the metaphorical line of 'below three' in language. Intersemiotic Polysemy operates in the co-contextualizing relations to construct semantic convergence between language and gesture. Prominence is given to the standard of 'below three' indicated by the line embodied in gestural selection. The focus on not meeting the standards is emphasized through the gestural repetition of the line as well as in the speed and force in which the action is performed. This prominence is also consistent with the language choice, where the category of students 'who score below three' is foregrounded as a marked Theme in the textual meaning.

Re-contextualizing relations between language and gesture constructs semantic divergence. However, these relations can be reconciled through the operation of Intersemiotic Polysemy. The emergent meaning usually has new semantic layers of sarcasm or ambivalence. An instance of this is observed

Figure 10. Offering Gesture.

when Andy makes the Representing Action of pretending to take out a tissue article from his pocket to offer a student. While the meaning of the Representing Action is an offer, this re-contextualizes with the linguistic texts where he says, 'How can you go to the toilet in the first few minutes of the lesson? You need a tissue also?' (see Figure 10). The gestural selection indicates overt support for the student to visit the toilet. However, the questions in language cast aspersions on the student's request to go to the toilet when the lesson has barely started. The operation of Intersemiotic Polysemy reconciles the semantic divergence and produces a new semantic layer of sarcasm in the emergent meaning.

CONCLUSION

Investigating intersemiosis is the core of the SFMDA approach. This article focuses, in particular, on the relationship between language and gesture made by teachers in the classroom and examines the emergent meanings made in their combination. It argues that the ideas developed for language–image relations are useful to provide a language for the description and discussion of the intersemiosis between language and gesture. In particular, Lim's (2004) formulation of co-contextualizing and re-contextualizing relations for language and images is applied to account for the semantic convergence

and semantic divergence between language and gesture. Liu and O'Halloran's (2009) mechanisms of Intersemiotic Parallelism and Intersemiotic Polysemy are also extended to describe the operations of intersemiotic cohesion in language and gesture. This is despite the recognition that images and gestures have different affordances and meaning potential. If the intersemiotic mechanisms can be productively applied, then the study, while limited to the few examples analysed from the classroom corpus, can tentatively point to a plausible case that at a certain level of abstraction, the principles of intersemiotic organization can be generalizable across semiotic resources. This conjecture presents possibilities for further exploration towards a unifying theory of intersemiotic organization across all semiotic resources.

While eliciting insights into the specific intersemiotic systems of language and gesture, the SFMDA approach is limited in that only two of the many multimodal resources, such as gaze, facial expression, use of positioning and movement in the teacher's multimodal ensemble, are studied. Other approaches, such as the multimodal (inter)action analysis and conversation analysis approach to embodied interactions, account for a wider range of semiotic resources in multimodal discourse analysis, albeit with a lesser focus on a fine-grained analysis of the specific features of each semiotic resource and the intersemiotic mechanisms between the resources in specific instantiations. Jewitt et al. (2016: 109) describe various approaches to multimodality and observe that different frameworks 'draw attention to and render visible different features of multimodal meaning making'. The SFMDA approach adopted in this article hopes to offer a perspective on how the intersemiosis between the teacher's use of language and gesture can be systematized and analysed from a systemic functional multimodal discourse analysis approach. While there have been studies that have examined the nature of gesture and the relationship between gesture and language in classroom interactions, this article focuses on the 'multimodal grammatics' in the contextualizing relations and intersemiotic mechanisms to bring about semiotic cohesion. As a theoretical contribution, it also sheds light on the ways in which multimodal semiotic resources work together across contexts.

Understanding the intersemiosis between the teachers' use of language and gesture in the classroom context has implications for pedagogy and teachers' professional development. From the overall analysis of the multimodal choices made by Andy and Betty in their lessons discussed in this article, as well as in the larger study, several key observations and interpretations were made. Betty shows a clear structure in the sequencing of the stages in her lesson, and has limited her positioning and movement to the front of the classroom. However, a sense of casualness and informality is injected through the meanings made through language, such as the high use of modality and adjuncts to construct solidarity with the students, as well as through gestures, such as the use of Indexical Action to realize openness. Through the orchestration of semiotic resources, Betty achieves her lesson objectives, encourages

students' participation and brings about overall enjoyment of her lesson, as signalled by the frequent occurrences of her students' laughter. In contrast, Andy's lesson is less well structured. He moves around the classroom a lot more than Betty during the lesson, and tends to position himself front off-centre of the classroom. Overall, Andy's use of language and gesture also realizes more mixed meanings with an extreme range – from casualness to authorativeness. This may be due to his attempt to 'control' his class, when he feels that they are too disorderly. The emergent meaning of 'Structured informality', as observed in Betty's lesson, provides insights into how an effective learning experience can be designed by the teacher through the orchestration of multimodal semiotic resources.

In structured informality, explicit displays of power dynamics between the teacher and students are managed. Through specific semiotic choices that function to maintain a didactic structure for learning, other semiotic choices are made to mitigate the hierarchical distance between the teacher and students (Lim et al., 2012). This achieves a degree of rapport between teacher and students and realizes a more participative pedagogy. Structured informality in the classroom is proposed as a way for teachers to design a collegial learning experience, within a structured unfolding of the lesson, and where students are encouraged to participate actively in their learning. It invites teachers to consider how their meaningful use of multimodal semiotic resources can design for participative learning.

ACKNOWLEDGMENTS

The author wishes to thank Yvonne Gan for the illustration of the figures.

FUNDING

This work was supported by the Singapore National Research Foundation's Interactive Digital Media R&D Programme [Grant Number NRF 2007 IDMIDM 002 - 066].

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